# Machine Learning Roadmap

## Stage 1 — Foundations (Don’t skip!)

- Python Essentials: variables, loops, functions, lists, dicts, sets, file handling, NumPy, Pandas, Matplotlib, Seaborn.
- Math for ML: Linear Algebra (vectors, matrices), Probability & Statistics, Calculus basics.
- Data Visualization: scatter plots, histograms, box plots, heatmaps.

## Stage 2 — Machine Learning Basics

- Core ML Concepts: features (X) & labels (y), train/test split, overfitting vs underfitting, evaluation metrics.
- Algorithms: Linear Regression, Logistic Regression, Decision Trees, KNN, Naive Bayes, SVM.
- Practice: sklearn.datasets, build pipelines in scikit-learn.

## Stage 3 — Intermediate Skills

- Data Preprocessing: handle missing data, encoding categorical variables, feature scaling.
- Model Optimization: cross-validation, hyperparameter tuning.
- Feature Engineering: create new features, feature selection.

## Stage 4 — Advanced ML

- Unsupervised Learning: KMeans, DBSCAN, Hierarchical Clustering, PCA, t-SNE.
- Ensemble Methods: Random Forest, Gradient Boosting, Bagging & Boosting.
- Model Deployment: save models (joblib, pickle), deploy with Flask/FastAPI or Streamlit.

## Stage 5 — Specialized Areas

- Natural Language Processing (NLP): tokenization, stopwords, stemming, lemmatization, embeddings, transformers.
- Deep Learning: neural networks, CNNs, RNNs/LSTMs.
- Recommendation Systems: collaborative & content-based filtering.

## Stage 6 — Projects & Portfolio

- House price prediction, spam classifier, customer segmentation, sentiment analysis, image classification.
- Upload to GitHub, write short blogs/notes.

## Stage 7 — Continuous Learning

- Join Kaggle competitions, read research papers, watch ML talks, explore new libraries.